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PROJECT PROPOSAL

DEVELOPMENT OF A SYSTEM FOR FAST VIDEO DECODING IN AV1 FORMAT

Bersenyov Ilya, ISD-22-2

Supervisor: Senior lecturer, Scientific director of the Department of Information Technology in Business, Academic director of the educational program development of information systems for business Lanin Vyacheslav Vladimirovich

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This work addresses the problem of optimizing the video decoding process for the AV1 format, specifically for the company Comexp LLC. In the current configuration of the TAPe video indexing system, decoding time accounts for over 95% of the total processing time. The project aims to develop and experimentally evaluate algorithmic optimization methods for the AV1 decoder in a single-threaded mode, operating at an ultra-low resolution and without utilizing CUDA acceleration. The research aims to identify decoder bottlenecks through a literature review, apply methods such as replacing lookup tables with analytical computations, optimize cache memory usage, and adapt post-processing algorithms for the target resolution. The expected outcome is a 30–50% reduction in decoding time within the final system, while maintaining the video quality required for the indexing process.

Introduction

Background.

The video encoding industry has traditionally been one of the most complex technical fields, demonstrating impressive results. For instance, the H.264 video codec, whose first standard version was released in 2003, achieves a video size reduction of 50–500 times compared to frames encoded sequentially as PNG images. An uncompressed 5-minute video at a resolution of 1920×1080, 30 frames per second, in 8-bit RGB representation would have a size of $1920 \times 1080 \times 3 \times 30 \times 300 \approx 5.6 \times 10^{10}$ bytes ($\approx$ 52 GB). Such remarkable efficiency is achieved through meticulously designed standards that account for modern processor architectures, mathematical optimizations in linear algebra, motion vector encoding technology, and methods for predicting subsequent frames. With each new codec standard, this efficiency continues to improve, as confirmed by a comparative analysis of different codec generations by Akyazi and Ebrahimi (2018).

However, this high compression efficiency has a downside: as compression ratios increase, so does the required computational load for decoding the file. This graduation thesis addresses a real business problem for the company Comexp LLC (https://comexp.net/). The company's primary product is the patented TAPe video indexer, which enables video-based video search. The core issue is that, in the current configuration, decoding time significantly exceeds indexing time. For example, decoding the two-hour movie "The Fate of the Furious" takes 247 seconds, while indexing it takes only ten seconds. Consequently, video indexing accounts for less than five percent of the total system processing time. For the company's clients, this means that when processing large volumes of video data (e.g., media archives or surveillance footage), the indexing speed becomes irrelevant, as the system spends 95% of its time on routine decoding. This creates a bottleneck, increasing the total cost of ownership and limiting the product's appeal to clients with real-time processing needs. These measurements were conducted on the company's server equipped with an Intel Xeon E5-2697 v2 processor, operating in single-threaded mode and a 200 GB SSD with an output resolution of 64px.

Problem statement.

The aim of this work is to develop and experimentally evaluate methods for optimizing AV1 decoding at a reduced output resolution (64px) in single-threaded mode. The objective is to engineer a system that enables rapid video upload and subsequent decoding while maintaining fault tolerance. The primary research focus is the algorithmic optimization of the AV1 decoder.

Delimitations of the study.

This work will examine a broad spectrum of technologies, including backend, frontend, and low-level programming. However, due to the high resource intensity of video decoding, the literature review will primarily focus on the topic of video decoder optimization. Decoder optimization will be performed for scaling videos to 64px while preserving the original aspect ratio, operating in single-threaded mode, and without utilizing CUDA technology. This approach is justified by several factors: single-threaded optimizations are more portable across different execution environments, are lightweight, can be deployed on client devices, and allow for isolating the effects of parallelism on performance measurements, thereby enabling a focused examination of algorithmic improvements. CUDA will not be employed at this stage, as its implementation is planned for future work and will build upon the optimizations developed in the current research. The ultra-low output resolution is dictated by the specific operational requirements of the TAPe system. The project will utilize the existing AV1 video codec implementation, Libaom, which conforms to the AV1 specification by Alliance for Open Media (2021), and will be executed in single-threaded mode targeting a generic architecture.

Professional significance.

The practical significance of this work stems from its direct relevance to the real-world business processes of Comexp LLC and its flagship product, the TAPe video indexing system. Accelerating AV1 video decoding, the primary objective of this research, will yield the following practical outcomes:

1. Reduced Equipment Downtime and Increased System Throughput. Decreasing the decoding time share from the current 95% to the target values will enable processing a larger volume of video material per unit of time without requiring additional computational resources.
2. Enhanced Efficiency of Internal Processes. Faster indexing directly impacts the speed of video data search, analysis, and processing, which is critically important for the tasks addressed by the company's clients.
3. Potential for Solution Scaling. The optimizations developed, being focused on single-threaded execution, can be subsequently adapted for embedded systems and resource-constrained client devices, and can also serve as a foundation for future implementations utilizing parallel computing technologies such as CUDA.

Thus, the results of this work will have direct practical application, manifesting as a measurable performance improvement in an existing commercial product.

Definitions of key terms.

Video Codec (Codec): A technology (standard and its software implementation) for compressing (encoding) and decompressing (decoding) digital video.

Video Decoder (Decoder): The component of a codec that decompresses a compressed video stream for playback or further processing.

TAPe: A patented video indexing system developed by Comexp LLC.

AOMedia Video 1 (AV1): An open video codec standard developed by the Alliance for Open Media (AOMedia) consortium.

Errata (AV1-v1.0.0errata): The specific standard version defining the behavior of the AV1 video codec.

Peak signal-to-noise ratio (PSNR): A metric to estimate the information loss in a video.

Libaom: An implementation of the AV1 standard by Aomedia.

Constrained directional enhancement filter (CDEF): in-loop filtering tool in the AV1 video codec that reduces ringing and artifacts.

Literature Review

The primary goal of this project is to identify methods for maximizing the optimization of video decoding at a low output resolution for the AV1 format, including the transmission pipeline from user to client.

To better understand AV1, it is instructive to compare it with the H.264 codec by examining the AV1 specification by Alliance for Open Media (2021) and the H.264 recommendation by ITU-T (2021). The H.264 standard encodes images using motion vector technology, primarily employing S, P, B, and I frames, as described in "THE H.264 ADVANCED VIDEO COMPRESSION STANDARD" book by Richardson (2010). Each frame is either a coded set of pixels or predicted from one or two neighboring frames. In AV1, the prediction system is significantly more complex, featuring a 10-level temporal structure. This means a single frame may depend on up to nine frames rather than two, substantially increasing decoding complexity. Additionally, the deblocking filter—a filter that reduces blocking artifacts—consumes considerable processing time. While this technology has remained largely unchanged in AV1, it remains resource-intensive. Understanding these aspects will help identify potential optimization targets.

For instance, a study by Ren and Kehtarnavaz (2007) demonstrated optimization of the adaptive deblocking filter operating on 4x4 block boundaries. Its computational complexity, despite relatively simple operations (shifts, additions), remains high due to the need to process every pixel at block boundaries. Unlike H.264, AV1 employs a more sophisticated multi-stage filtering approach, detailed in Chapter VII of another publication (Han et al., 2021), making direct replication of methods impossible. However, the key optimization concept—leveraging context information from the bitstream to adapt computational load—remains relevant for this work. Nevertheless, there is a risk of poor reproducibility, as the methods were applied to the JM9.5 reference decoder, which implements only the minimum required by the standard. This inherently leaves substantial room for optimization, and applying similar techniques to AV1 may not yield significant results.

Compared to its predecessor VP9, AV1 is considerably more complex and computationally expensive. While this reduces the final encoded video size, it also increases encoding and decoding time and complexity, as demonstrated in a study by Akyazi and Ebrahimi (2018).

Specifically, a paper by Chen et al. (2018) identifies the most resource-intensive components: coding block partitioning, interpolation, and the constrained directional enhancement filter. The coding block partition tree, compared to VP9, not only increases the initial block size (also called a superblock in AV1 and VP9 specifications) from 64×64 to 128×128 but also expands the number of possible partitions from four to ten, as discussed in Section II-A. Another highly resource-intensive component is the aforementioned constrained directional enhancement filter, mentioned in Section II-F-1. This nonlinear filter operates after the deblocking filter and requires computing edge direction for every 8×8 block in the frame, a computationally demanding operation. Additionally, the article discusses interpolation, highlighting its relationship with other operations. Compared to VP9, interpolation precision has increased eightfold, from 1/8 to 1/64. This means that calculating a single pixel using an 8-tap filter requires loading 8 pixels from memory and 8 coefficients, with 64 different coefficient sets divided into groups such as smooth, sharp, and others. During decoding, this results in billions of cache-inefficient operations. A limitation of this paper is its focus on compression efficiency, providing virtually no data on decoding time, CPU utilization, or cache memory usage. The complexity of AV1's block partitioning system is further illustrated by Guo et al. (2018), who demonstrate that block structure similarity is preserved across resolutions — a finding relevant to low-resolution decoding optimization.

A study on the CAVLC entropy decoder for H.264 (Yi & Song, 2008) is relevant because H.264, like AV1, VP9, and HEVC (H.265), inherits concepts from MPEG. Therefore, some H.264 optimization ideas may prove useful for AV1. Specifically, the article discusses optimization methods such as hardware acceleration and replacing lookup tables. While hardware acceleration through parallelism is not of interest here—since the decoder will operate in single-core mode—the second approach is highly relevant. The paper demonstrates that replacing numerous lookup tables with algebraic expressions is significantly faster, as it eliminates multiple calculations and cache misses associated with data access, while also reducing RAM footprint.

Furthermore, optimizing standard computations by replacing them with efficient bitwise operations is worthwhile, as explored in an article by Moroz and

Samotyy (2018). This work examines the classic division-by-constant method using a "magic constant" and its optimized variant, which operates 30% faster. While division is less common in video decoding and appears more frequently in encoding, this technique—despite its complexity—could significantly accelerate interpolation, which involves dividing the sum of products of filter coefficients and pixel values.

Existing solutions typically target either encoder optimization or high-resolution decoding optimization, as optimizing decoding at ultra-low resolutions is a rather niche and specific task. Nevertheless, these solutions employ interesting optimization concepts, some of which can be adapted for AV1 decoding with appropriate modifications.

Methods

This project will implement selected optimization techniques from the literature to enhance the Libaom video codec in single-threaded mode, while also incorporating various architectural optimizations beyond Libaom itself. These include low-latency databases, client-side file size reduction using FFMPEG compiled to WebAssembly, and the NATS Jetstream message broker.

System performance will be evaluated according to multiple criteria:

1. End-to-End Processing Time. The total time from user file upload to complete server-side decoding, measured against a naive implementation baseline.
2. Per-Frame Quality Loss. Computed as the difference in Luma values for each pixel between frames decoded by the reference Libaom decoder and its optimized version.
3. Per-Frame PSNR Metric. Calculation of the Peak Signal-to-Noise Ratio for each frame to quantify quality degradation.

4. Stability and Reproducibility. Validation of Libaom optimizations through a minimum of five test runs with identical input data to ensure consistent results.

5. System Response Under Load. Performance testing measuring response times with identical datasets, compared against the naive implementation.

To evaluate overall system optimization, a naive implementation will be developed using Flask and the unoptimized Libaom video codec, replicating the functionality of the optimized system while serving as a baseline for comparison.

Results Anticipated

Upon completion of this work, the following outcomes are anticipated:

1. Identification of AV1 Decoding Bottlenecks in single-threaded mode at 64px output resolution, including:
   a. Computational complexity of filtering operations,
   b. Interpolation processes,
   c. Entropy decoding,
   d. Architectural component overhead within the overall system.
2. Development of a Modified Libaom Decoder Version incorporating:
   a. Reduced memory access operations,
   b. Replacement of lookup tables with analytical computations,
   c. Optimization of arithmetic operations,
   d. Adaptation of post-processing for low output resolution.
3. Experimental Validation of Performance Improvement:
   a. At least 20–35% acceleration at the decoder level,
   b. At least 30–50% acceleration at the overall system level compared to the naive implementation.

4. Confirmation of Acceptable Quality Degradation across various video types, including static films (e.g., works by Andrei Tarkovsky) and dynamic films with numerous scenes (e.g., Avengers, Fast & Furious):
   a. PSNR ≥ acceptable threshold of 15 dB,
   b. Absence of critical artifacts affecting TAPe functionality.

It is worth noting that to search using tape, you do not need high-quality video, but you need it to be uniquely decoded into a sequence of frames in order to distinguish the videos from each other. This justifies the low PSNR value.

Conclusion

This work has formulated and substantiated the problem of optimizing AV1 video decoding under conditions of ultra-low output resolution (64px) in single-threaded mode, as applied to the real-world production system TAPe at Comexp LLC. Analysis of the current system state has revealed a critical imbalance: decoding accounts for over 95% of the total processing time, making it an obvious and priority target for optimization.

The literature review has identified the most resource-intensive components of the AV1 decoder and determined promising optimization methods—ranging from replacing lookup tables with analytical computations to adapting post-processing for the target resolution. The combination of planned algorithmic and architectural optimizations provides a reasonable basis for expecting a 30–50% system acceleration compared to the naive implementation, while maintaining acceptable decoding quality.

The results of this work will have direct practical application within the company's product.

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